

AI Idea Forge – QA & Security Review Report

Review Status: Pre-Presentation Quality Assurance

A comprehensive QA and Security Review was conducted prior to project completion. High-priority issues affecting security, stability, and data integrity were resolved before release, while medium and low-priority improvements were documented for future iterations.

Issue #2 – File Size & MIME Validation

Description: The application accepted uploaded files without validating file size or MIME type despite displaying a 10 MB limit.

Risk: Unsupported or oversized files could enter the workflow, causing failures and inconsistent AI processing.

Resolution: Implemented file type validation and enforced the 10 MB upload limit with user-friendly validation messages.

Outcome: Only supported file types within the allowed size can now be uploaded.

Issue #3 – Website URL Validation

Description: Any non-empty text was accepted as a website URL.

Risk: Invalid URLs could be sent to downstream AI workflows, reducing analysis quality.

Resolution: Added validation to accept only valid HTTP/HTTPS URLs.

Outcome: Only valid URLs are submitted for analysis.

Issue #11 – Duplicate / Disconnected Records

Description: Potential inconsistencies were identified between analysis records, activity logs, and workflow identifiers.

Risk: Could result in orphaned records and inconsistent reporting.

Resolution: Reviewed the architecture and ensured consistent identifier usage across Supabase, n8n, and Discovery Galaxy.

Outcome: Related records now maintain consistent relationships.

Issue #12 – Duplicate Submission Prevention

Description: Retrying failed submissions could create duplicate analyses and AI executions.

Risk: Duplicate records increase processing cost and confuse users.

Resolution: Improved retry handling to prevent duplicate submissions.

Outcome: Retries no longer generate duplicate analyses.

Issue #17 – Polling Lifecycle Optimization

Description: Background polling continued after page navigation.

Risk: Unnecessary database reads and potential memory leaks.

Resolution: Optimized polling lifecycle and cleaned up background tasks.

Outcome: Improved performance and reduced unnecessary database activity.

Issue #22 – Report Download Error Handling

Description: PDF generation failures were silently ignored.

Risk: Users received no feedback when downloads failed.

Resolution: Added exception handling with user-friendly notifications.

Outcome: Users now receive clear feedback when report generation fails.

Issue #24 – Authentication Route Protection

Description: Protected pages were accessible without proper authentication checks.

Risk: Unauthenticated users could reach protected screens and experience inconsistent behaviour.

Resolution: Implemented route guards to redirect unauthenticated users to the login page.

Outcome: Protected pages now require authentication.

Conclusion

The QA review identified multiple opportunities to improve reliability, security, and user experience. Critical and High-priority findings were resolved before the final presentation, while remaining enhancements were documented for future releases.